

MATH 220/199 Week 13 Day 1

Reminders

1. Office hours are held in Davenport 330 at 3pm on Thursdays.
2. Do the Merit Quiz on the Merit Canvas page.

What if...Not Rectangle?

One issue with estimating areas with rectangles is that if a function is very concave/convex, we will get very large overestimates/underestimates. What if we used a new shape, like a trapezoid?

1. Let $f(x) = x^2$. Fill in the chart below:

x	0	1	2	3	4
$f(x)$					

2. Compute a Left Riemann sum for the area between $f(x)$ and the x-axis using 4 rectangles.
3. Compute a Right Riemann sum for the area between $f(x)$ and the x-axis using 4 rectangles.
4. Now, compute the 4-trapezoidal sum: Construct the trapezoids the same way as the rectangle, but the top side is made by connecting $f(a)$ to $f(a + \Delta)$.
5. Using $\int_0^4 x^2 \, dx = \frac{64}{3}$, which estimate is the best?

Exploring Summation Notation

The following problems only look hard, they are not actually hard. My goal is to convince you that summation notation is not scary.

1. Let $f^{(n)}(x)$ denote the n^{th} derivative of $f(x)$. Say $f^0(x) := f(x)$. Let $f(x) = \sin(x)$. Compute:

$$\sum_{i=0}^8 f^{(2i)}\left(\frac{\pi}{2} + i\pi\right)$$

2. Compute $\sum_{n=0}^{10} (-1)^n \cdot n$

3. Let $\Delta_{(n,m)}$ be the right triangle with leg lengths n, m . Define $\mathbb{A}(\Delta_{(n,m)}) := \text{Area of } \Delta_{(n,m)}$. Compute:

$$\sum_{n=1}^4 \mathbb{A}(\Delta_{(n,n+1)})$$